Kavach & Electronic Interlocking (EI) – SIP Symbol Register

S.No. | Symbol / notation | Meaning and application

Reference register for engineering review and AI drawing classification. The notation column is a searchable shorthand, NOT a replacement for the approved graphical symbol. Original IRISET SIP symbol plates are reproduced in the appendix. Kavach tag letters are from RDSO Annexure H, Amendment 5. Confirm project-specific legends and the current approved standards before design use.

A. Electronic Interlocking / Signal Interlocking Plan

S.No.	Symbol / notation	Meaning
1	2A CL	Two-aspect colour-light signal; signal profile and permissible indications depend on approved signalling arrangement.
2	3A CL	Three-aspect colour-light signal; verify lamp positions and aspects from signal profile.
3	4A CL	Four-aspect colour-light signal; double yellow is an aspect, not an additional signal type.
4	HOME	Home signal: controls entry to station limits / reception routes.
5	STARTER	Starter signal: controls departure from a station line.
6	ADV STARTER	Advanced starter: controls departure toward block section.
7	DISTANT	Distant signal: advance warning of next stop signal.
8	IBS	Intermediate block stop signal.
9	AUTO / A	Automatic stop signal with automatic marker plate.
10	SEMI-AUTO	Semi-automatic signal with applicable identification.
11	CALL-ON	Calling-on subsidiary signal for restricted-authority movement.
12	SHUNT	Shunt signal: controls shunting movements.
13	PL SHUNT	Position-light shunt signal.
14	REPEATER	Repeater signal indicating aspect of signal ahead.
15	ROUTE IND	Route indicator; junction / theatre / LED matrix as applicable.
16	CO-ACT	Co-acting signal for sighting where provided.
17	GATE SIG	Gate stop signal associated with level crossing.
18	STOP BOARD	Stop board / stop marker.
19	BSLB	Block section limit board.
20	S / SIGNAL ID	Signal identification / numbering shown next to symbol.
21	P-N	Point / turnout in normal position.
22	P-R	Point / turnout in reverse position.
23	X-OVER	Crossover pair of points.
24	TRAP	Trap point / derailing switch.
25	FPL	Facing point lock where relevant.
26	FM	Fouling mark.
27	IRJ	Insulated rail joint.
28	TC	Track circuit / track detection section.
29	AC	Axle counter section.
30	DP	Axle counter detection point / wheel sensor location.
31	OVL	Overlap / overlap limit.
32	CSR	Clear standing room.
33	LC	Level crossing gate and associated apparatus.
34	CH	Crank handle / point emergency operating handle.
35	SPT	Signal post telephone.
36	LOC	Location box / apparatus case.
37	RR / EI	Relay room / electronic interlocking equipment building.
38	GF	Ground frame.
39	BUFFER	Buffer stop / dead end.
40	SAND HUMP	Sand hump / overrun protection.

41	GRAD	Gradient / gradient post.
42	KM	Kilometre post / chainage.
43	UP / DN	Up and Down track direction annotation.
44	PLATFORM	Platform / station infrastructure.
45	BRIDGE	Bridge / tunnel reference where signalling-relevant.
46	NEW / RED	New equipment shown red in scheme plans.
47	REMOVE / GREEN	Equipment to be removed shown green in scheme plans.
48	SOURCE / BLUE	Addition sourced outside base drawing shown blue in scheme plans.

Basis: IRISET Standardisation of S&T Drawings V2.1, Chapter 6, pages 98–105; signal definitions supplemented by general engineering terminology.

B. Kavach – RFID Tag / TIN Layout

S.No.	Symbol / notation	Meaning
1	N	Inline section RFID tag.
2	S	Signal-foot RFID tag.
3	T	TIN discrimination / turnout RFID tag.
4	X	Kavach territory exit RFID tag.
5	D	Dead-stop RFID tag.
6	G	Level crossing gate RFID tag.
7	L	Adjacent-line information RFID tag.
8	A	Adjustment / junction RFID tag.
9	TIN	Track Identification Number, denoting the track section in Kavach layout.
10	TAG ID	Unique RFID tag number shown on the layout.
11	MAIN + DUP	Main and duplicate RFID tags, subject to prescribed placement.
12	NOM / REV	Nominal and reverse travel directions.
13	SK ID	Stationary Kavach identity shown with station name.
14	RFID CHAINAGE	Absolute location / chainage of a tag.
15	TIN COLOUR	Distinct colour used for adjacent TIN sections; not a fixed aspect colour.
16	WHITE	Non-Kavach territory, shown white in TIN layout.
17	RTU	Remote terminal unit where represented in project layout.
18	RADIO	Radio network / antenna or tower location: project-specific legend required.
19	STATION KAVACH	Stationary Kavach equipment location: project-specific legend required.
20	LOCO KAVACH	Onboard Kavach equipment; normally in architecture, not a standard SIP track symbol.

Basis: RDSO Kavach RFID Tag-TIN Layout Guidelines, Annexure H, Amdt-5 (especially tag notation table); supplementary equipment descriptions are project-dependent.

Reference notes

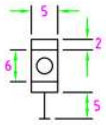
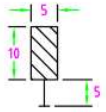
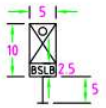
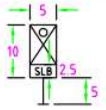
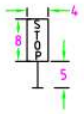
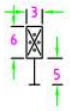
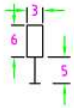
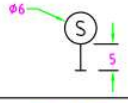
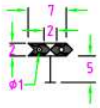
- The 2/3/4-aspect entries are classifications. Their aspect colours (R/Y/YY/G) must be checked against the approved aspect sequence chart and signal profile.
- IRISET page 23 states that the normally displayed aspect is drawn with a double line; graphical shapes in the appendix are the actual source symbols.
- Kavach Annexure H specifies letters N, S, T, X, D, G, L, A for tag placements. It does not establish a universal pictorial icon for every Kavach equipment item.
- Sources: IRISET https://iriset.indianrailways.gov.in/instt/uploads/files/1581487978275-StdOfSnTDrawingsV2.1.pdf ; RDSO https://rdso.indianrailways.gov.in/uploads/files/Annexure-H%20-KAVACH%20RFID%20Tag-TIN%20Layout%20guidelines%20Admt-5_0.pdf

Appendix – Original IRISET SIP symbol plates (pages 99–105)

Original IRISET page 99

STANDARD SYMBOLS FOR SIGNAL INTERLOCKING AND SCHEME PLANS

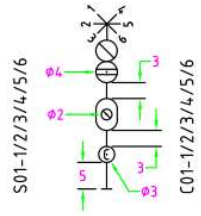
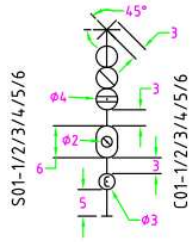
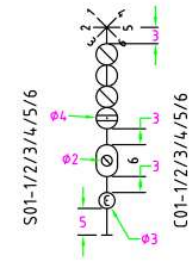
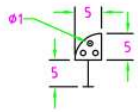
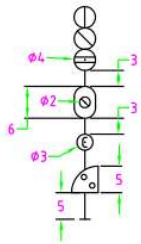
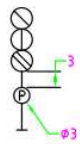
(ALL DIMENSIONS IN MM)

SNO.	FUNCTION	SYMBOL	DESCRIPTION
1.	BOARD		WARNING BOARD (GOODS)
2.	BOARD		WARNING BOARD (PASSENGER)
3.	BOARD		BLOCK SECTION LIMIT BOARD (BSLB)
4.	BOARD		SHUNTING LIMIT BOARD (SLB)
5.	BOARD		STOP BOARD
6.	BOARD		LEGEND BOARD
7.	BOARD		CALLING ON LEGEND BOARD
8.	BOARD		GENERAL PURPOSE LEGEND BOARD
9.	BOARD		MID SECTION SIDING BOARD ('S' MARKER)
10.	BOARD		MID SECTION SIDING CAUTION INDICATOR
11.	BOARD		SPEED INDICATOR

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STANDARD SYMBOLS FOR SIGNAL INTERLOCKING AND SCHEME PLANS

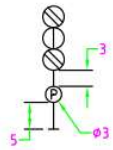
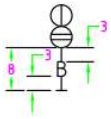
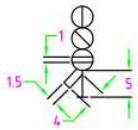
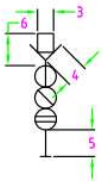
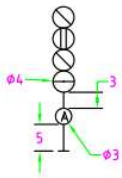
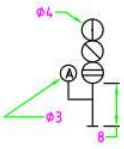
(ALL DIMENSIONS IN MM)

SNO.	FUNCTION	SYMBOL	DESCRIPTION
12.	SIGNAL		2-ASPECT COLOUR LIGHT SIGNAL (HOME) WITH ROUTE INDICATOR/CALLING ON C MARKER BOARD
13.	SIGNAL		3-ASPECT COLOUR LIGHT SIGNAL (HOME) WITH ROUTE INDICATOR/CALLING ON 'C' MARKER BOARD
14.	SIGNAL		4-ASPECT COLOUR LIGHT SIGNAL (HOME) WITH ROUTE INDICATOR/CALLING ON C MARKER BOARD
15.	SIGNAL		COLOUR LIGHT SHUNT SIGNAL (INDEPENDENT)
16.	SIGNAL		MAIN WITH CALLING-ON & SHUNT SIGNAL (DEPENDENT)
17.	SIGNAL		INNER DISTANT SIGNAL

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STANDARD SYMBOLS FOR SIGNAL INTERLOCKING AND SCHEME PLANS

(ALL DIMENSIONS IN MM)

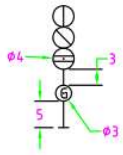
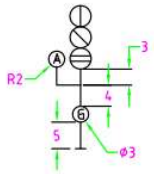
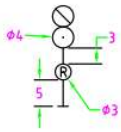
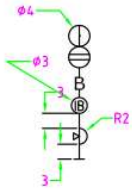
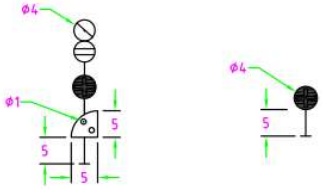
SNO.	FUNCTION	SYMBOL	DESCRIPTION
18.	SIGNAL		DISTANT SIGNAL
19.	SIGNAL		ADV. STR. CONTROLLED BY BLOCK INSTRUMENT
20.	SIGNAL		SLOTTED SIGNAL (SLOT)
21.	SIGNAL		3-ASPECT CLS WITH LED MATRIX WITH DIRECTION TYPE ROUTE INDICATOR
22.	SIGNAL		4-ASPECT COLOUR LIGHT AUTO SIGNAL
23.	SIGNAL		3-ASPECT COLOUR LIGHT SEMI AUTO SIGNAL

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STANDARD SYMBOLS FOR SIGNAL INTERLOCKING AND SCHEME PLANS

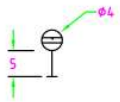
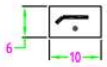
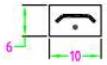
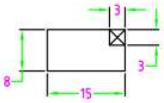
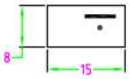
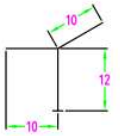
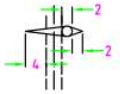
(ALL DIMENSIONS IN MM)

SNO.	FUNCTION	SYMBOL	DESCRIPTION
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25.	SIGNAL		3-ASPECT COLOUR LIGH SEMI-AUTOMATIC GATE SIGNAL
26.	SIGNAL		COLOUR LIGHT REPEATER SIGNAL
27.	SIGNAL		COLOUR LIGHT STARTER INDICATOR SIGNAL
28.	SIGNAL		COLOUR LIGHT I.B. SIGNAL PROVIDED WITH SIGNAL POST TELEPHONE
29.	SIGNAL		SHUNT PERMITTING INDICATOR

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STANDARD SYMBOLS FOR SIGNAL INTERLOCKING AND SCHEME PLANS

(ALL DIMENSIONS IN MM)

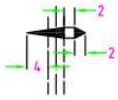
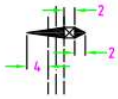
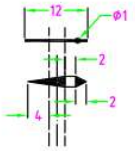
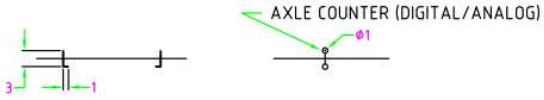
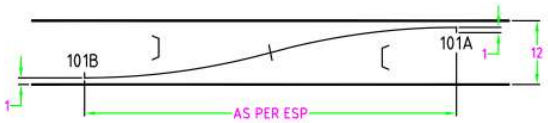
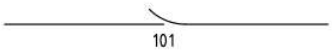
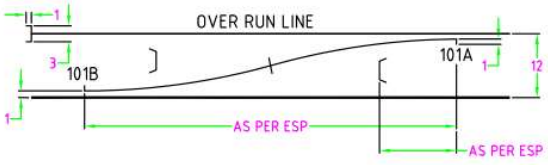
SNO.	FUNCTION	SYMBOL	DESCRIPTION
30.	OPERATION PLACE		FIXED RED SIGNAL
31.	OPERATION PLACE		(S.M.) PANEL WITH VDU
32.	OPERATION PLACE		(S.M.) DUAL VDU
33.	OPERATION PLACE		GATE GOOMTY (WINCH OPERATION)
34.	OPERATION PLACE		GATE GOOMTY (PANEL OPERATION)
35.	GRADIENT		GRADIENT
36.	ARC LEVER FOR MECH. POINT OPERATION		GROUND LEVER FRAME WITH SINGLE LEVER
37.	INDICATOR		KEY LOCK
38.	INDICATOR		INDICATOR (POINT/TRAP)
39.	L.C. GATE		NON INTERLOCKED LC. GATE

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STANDARD SYMBOLS FOR SIGNAL INTERLOCKING AND SCHEME PLANS

(ALL DIMENSIONS IN MM)

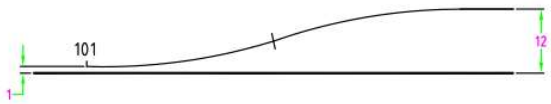
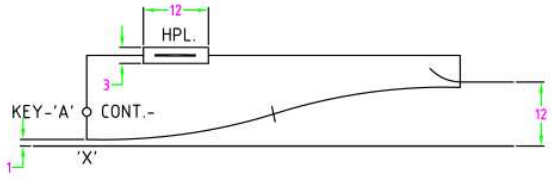
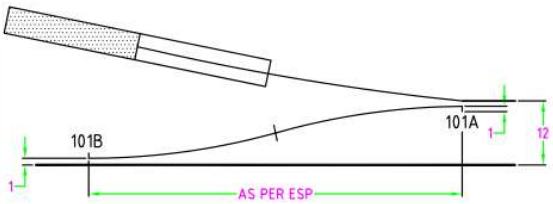
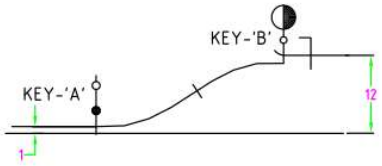
SNO.	FUNCTION	SYMBOL	DESCRIPTION
40.	L.C. GATE		INTERLOCKED LC. GATE (MOTOR OPERATED)
41.	L.C. GATE		INTERLOCKED LC. GATE (MECHANICAL OPERATED)
42.	L.C. GATE AND SLIDING BOOM		INTERLOCKED LC. GATE WITH SLIDING BOOM ATTACHED WITH KEY
43.	DIRECTION OF SIDE		DIRECTION NORTH LINE
44.	DIRECTION OF SIDE		DIRECTION OF TRAFFIC (UP & DN)
45.	LINES		GAP BETWEEN ADJACENT LINE
46.	TRACK CIRCUIT/ AXLE COUNTER		TRACK CIRCUIT (DC/AXLE COUNTER)
47.	POINT		X-OVER POINT SHOWING FM.
48.	POINT		TRAP POINT (DERAILING SWITCH)
49.	POINT		X-OVER POINT SHOWING FM., OVER RUN LINE AND DEAD END

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STANDARD SYMBOLS FOR SIGNAL INTERLOCKING AND SCHEME PLANS

(ALL DIMENSIONS IN MM)

SNO.	FUNCTION	SYMBOL	DESCRIPTION
50.	POINT		POINT (SINGLE END)
51.	POINT		CONTROL POINT (ELECTRICALLY KEY CONTROL OPERATED BY HAND POINT LEVER)
52.	POINT		X-OVER POINT (SHOWING SAND HUMP)
53.	POINT		CONTROL POINT OPERATED BY HP LEVER WITH SUCCESSIVE ARRANGEMENT KEY CONT.
54.	LINE		BREAK LINE
55.	LINE		CENTRE LINE OF STATION BULDING
56.			ELECTRICAL KEY TRANSMITTER (EKT/RKT) PAGE NO.-7/7